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DMA025A

Mathematics-II

Department of Mathematics
JECRC University Jaipur



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Unit-V

Complex Variable–Integration





DMA025A	Mathematics-II	3+1+0 = 4 Credits
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Course Contents:

UNIT I: Matrices:

Linear Systems of Equations, Linear Independence, Rank of a Matrix; Determinant, Inverse of a matrix, rank-nullity theorem, System of linear equations, Symmetric, skew-symmetric and orthogonal matrices, Determinants, Eigen values and eigenvectors, Orthogonal transformation, Diagonalization of matrices, Cayley-Hamilton Theorem.

UNIT II: First order ordinary differential equations:

Exact, linear and Bernoulli's equations. Equations not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's type.

UNIT III: Ordinary differential equations of higher orders:

Second order linear differential equations with variable coefficients: Euler-Cauchy equations, solution by variation of parameters; Power series solutions: Legendre's equations and Legendre polynomials, Frobenius method, Bessel's equation and Bessel's functions of the first kind and their properties.

UNIT IV: Complex Variable–Differentiation:

Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate, elementary analytic functions(exponential, trigonometric, logarithm) and their properties, Conformal mappings, Mobius transformations and their properties.

UNIT V: Complex Variable–Integration:

Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour.

Text/Reference Books:



- AICTE's Prescribed Textbook: Mathematics-II (Calculus, Ordinary Differential Equations and Complex Variable), Khanna Book Publishing Co.
- Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
- Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.
- Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, John Wiley & Sons, 2006.
- Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.



Course Outcome (CO):



Upon successful completion of this course, the student will be able to:

- CO1: The students will learn the essential tool of matrices and linear algebra in a comprehensive manner.
- CO2: Understand the linear ordinary differential equations along with their applications.
- CO3: Will able to solve problem involving ordinary differential equations of higher orders.
- CO4: The students will learn the Complex Variable – Differentiation.
- CO5: The students will learn the Complex Variable–Integration.



Mapping of COs with Pos



MAPPING COURSE OUTCOMES LEADING TO THE ACHIEVEMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

Course Outcome	Program Outcome												Program Specific Outcome		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	M	L			M		M					M		M	
CO2	M	M		L	M		M			M	M	M	M		
CO3		L			M					L		M	M	M	
CO4	M	M		M	L		M					M		M	
CO5	M			M	M		H					M	M	M	

H=Highly Related ;M=Medium L=Low



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Contents



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- Line Integral.
- Contour integrals: Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof).
- Liouville's theorem and Maximum-Modulus theorem (without proof).
- Zeros of analytic functions, singularities.



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Introduction to Line Integral



In complex analysis, integration is performed along a curve in the complex plane. Such integrals are called **line integrals** or **contour integrals**.

Let

$$z = x + iy$$

Let a complex function be

$$f(z) = u(x, y) + iv(x, y)$$

Let C be a curve joining points A and B in the complex plane. The integral of $f(z)$ along the curve C is written as

$$\int_C f(z) dz$$

The value of the integral generally depends upon the path followed between the two points.



Definition of Line Integral



Let the curve C be represented parametrically by

$$z = z(t), \quad a \leq t \leq b$$

Then

$$dz = \frac{dz}{dt} dt$$

Hence the line integral becomes

$$\int_C f(z) dz = \int_a^b f(z(t)) \frac{dz}{dt} dt$$

If

$$f(z) = u + iv \quad \text{and} \quad dz = dx + i dy$$

then

$$\int_C f(z) dz = \int_C (u + iv)(dx + i dy)$$



Expansion of Line Integral



Expanding the expression

$$(u + iv)(dx + i dy)$$

we obtain

$$u dx - v dy + i(v dx + u dy)$$

Thus the complex integral can be written as

$$\int_C f(z) dz = \int_C (u dx - v dy) + i \int_C (v dx + u dy)$$

Hence evaluation of a complex integral reduces to the evaluation of two real integrals.



Example 1



Evaluate the following integral along the straight line joining (0, 0) and (3, 9) where $y = 3x$:

$$I = \int_C (x + iy)^2 dz.$$

Since the path is $y = 3x$, we substitute

$$z = x + i(3x) = x + 3ix$$

Hence

$$dz = dx + i dy$$

But $y = 3x \Rightarrow dy = 3dx$

$$dz = dx + 3i dx = (1 + 3i)dx$$

Now

$$(x + iy)^2 = (x + i(3x))^2 = x^2(1 + 3i)^2$$

$$(1 + 3i)^2 = 1 + 6i + 9i^2 = -8 + 6i$$

Thus

$$(x + iy)^2 = x^2(-8 + 6i)$$

Therefore

$$I = \int_0^3 x^2(-8 + 6i)(1 + 3i) dx$$

$$(-8 + 6i)(1 + 3i) = -26 - 18i$$



Contd...



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Hence

$$I = \int_0^3 x^2(-26 - 18i) dx$$

$$= (-26 - 18i) \int_0^3 x^2 dx$$

$$= (-26 - 18i) \left[\frac{x^3}{3} \right]_0^3$$

$$= (-26 - 18i) \times 9$$

$$I = -234 - 162i$$



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Example 2



Let's evaluate

$$\int_C \bar{z} dz$$

where C is given by $x = 3t$, $y = t^2$, with $-1 \leq t \leq 4$.

Here we have that C is

$$z(t) = 3t + it^2.$$

Therefore, with the identification $f(z) = \bar{z}$ we have

$$f(z(t)) = \overline{3t + it^2} = 3t - it^2.$$

Also,

$$z'(t) = 3 + 2it,$$

and so the integral is

$$\int_C \bar{z} dz = \int_{-1}^4 (3t - it^2)(3 + 2it) dt$$



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$$\begin{aligned} &= \int_{-1}^4 (2t^3 + 9t + 3t^2i) dt \\ &= \int_{-1}^4 (2t^3 + 9t) dt + i \int_{-1}^4 3t^2 dt \\ &= \left(\frac{1}{2}t^4 + \frac{9}{2}t^2 \right)_{-1}^4 + i t^3 \Big|_{-1}^4 \\ &= 195 + 65i. \end{aligned}$$



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Example 3



Now let's evaluate

$$\int_C \frac{1}{z} dz$$

where C is the circle $x = \cos t$, $y = \sin t$, with $0 \leq t \leq 2\pi$.

The parametric form of the curve is

$$z(t) = x + iy = \cos t + i \sin t = e^{it}.$$

Thus,

$$\frac{1}{z} = \frac{1}{e^{it}} = e^{-it}.$$

Also,

$$z'(t) = \frac{dz}{dt} = ie^{it}.$$

Substituting in the integral,

$$\int_C \frac{1}{z} dz = \int_0^{2\pi} e^{-it}(ie^{it}) dt = \int_0^{2\pi} i dt = 2\pi i.$$



Example 4



Evaluate $\int_0^{2+i} (\bar{z})^2 dz$ along the real axis from $z = 0$ to $z = 2$ and then along a line parallel to y -axis from $z = 2$ to $z = 2 + i$.

Solution. $\int_0^{2+i} (\bar{z})^2 dz = \int_0^{2+i} (x - iy)^2 (dx + i dy)$

$$= \int_{OA} (x^2) dx + \int_{AB} (2 - iy)^2 i dy$$

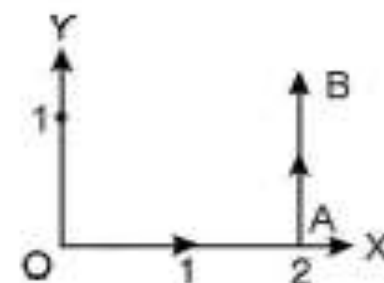
[Along OA , $y = 0$, $dy = 0$, x varies 0 to 2.

Along AB , $x = 2$, $dx = 0$ and y varies 0 to 1

$$= \int_0^2 x^2 dx + \int_0^1 (2 - iy)^2 i dy$$

$$= \left[\frac{x^3}{3} \right]_0^2 + i \int_0^1 (4 - 4iy - y^2) dy = \frac{8}{3} + i \left[4y - 2iy^2 - \frac{y^3}{3} \right]_0^1$$

$$= \frac{8}{3} + i \left[4 - 2i - \frac{1}{3} \right] = \frac{8}{3} + \frac{i}{3} (11 - 6i) = \frac{1}{3} (8 + 11i + 6) = \frac{1}{3} (14 + 11i)$$



Which is the required value of the given integral.

Ans.

Example 5



Example 5. Evaluate $\int_0^{1+i} (x^2 - iy) dz$, along the path

(a) $y = x$ (R.G.P.V., Bhopal, III Semester, Dec. 2007)

(b) $y = x^2$.

Solution.

(a) Along the line $y = x$,

$$dy = dx \text{ so that } dz = dx + idy \Rightarrow dz = dx + i dx = (1 + i) dx$$

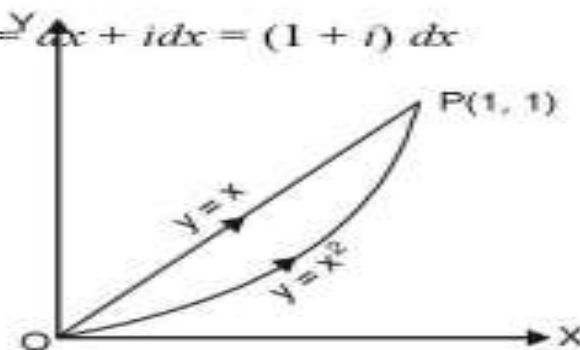
$$\therefore \int_0^{1+i} (x^2 - iy) dz$$

[On putting $y = x$ and $dz = (1 + i) dx$]

$$= \int_0^1 (x^2 - ix)(1 + i) dx$$

$$= (1 + i) \left[\frac{x^3}{3} - i \frac{x^2}{2} \right]_0^1 = (1 + i) \left(\frac{1}{3} - \frac{1}{2} i \right)$$

$$= \frac{(1 + i)(2 - 3i)}{6} = \frac{5}{6} - \frac{1}{6} i.$$



Which is the required value of the given integral.

Ans.

(b) Along the parabola $y = x^2$, $dy = 2x dx$ so that

$$dz = dx + idy \Rightarrow dz = dx + 2ix dx = (1 + 2ix) dx$$

x varies from 0 to 1.

$$\therefore \int_0^{1+i} (x^2 - iy) dx = \int_0^1 (x^2 - ix^2)(1 + 2ix) dx = \int_0^1 x^2 (1 - i)(1 + 2ix) dx$$

$$= (1 - i) \int_0^1 x^2 (1 + 2ix) dx = (1 - i) \left[\frac{x^3}{3} + i \frac{x^4}{2} \right]_0^1$$

$$= (1 - i) \left[\frac{1}{3} + \frac{1}{2} i \right] = \frac{(1 - i)(2 + 3i)}{6} = \frac{1}{6} (2 + 3i - 2i + 3) = \frac{5}{6} + \frac{1}{6} i$$

Which is the required value of the given integral.

Ans.

EXERCISE



1. Integrate $f(z) = x^2 + ixy$ from $A(1, 1)$ to $B(2, 8)$ along

(i) the straight line AB ; (ii) the curve $C, x = t, y = t^3$. **Ans.** (i) $-\frac{1}{3}(147 - 71)i$ (ii) $-\left(\frac{1094}{21} - \frac{124i}{5}\right)$

2. Evaluate $\int_{1-i}^{2+i} (2x + iy + 1) dz$ along

(i) $x = t + 1, y = 2t^2 - 1$; (ii) the straight line joining $1 - i$ and $2 + i$. **Ans.** (i) $4 + \frac{25}{3}i$ (ii) $4 + 8i$

(R.G.P.V., Bhopal, Dec. 2008)

3. Evaluate the line integral $\int_c (3y^2 dx + 2y dy)$, where c is the circle $x^2 + y^2 = 1$, counter clockwise from $(1, 0)$ to $(0, 1)$. **Ans. - 1.**



Contour Integrals



Let $f(z)$ be a complex function defined on a smooth curve C parameterized by

$$z = z(t), \quad a \leq t \leq b.$$

The contour integral of f along C is defined as:

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt.$$



Example



Evaluate

$$\int_C z^2 dz$$

where C is the line segment from 0 to $1 + i$.

Solution

Parameterize:

$$z(t) = t(1 + i), \quad 0 \leq t \leq 1.$$

Then

$$z'(t) = 1 + i$$





So,

$$\int_C z^2 dz = \int_0^1 (t(1+i))^2 (1+i) dt.$$

Since

$$(1+i)^2 = 2i,$$

we get

$$= \int_0^1 2it^2(1+i)dt.$$

Simplify and integrate.

Example. We evaluate $I_j = \int_{\gamma_j} x dz$, $j = 1$ to 7 , where





- (i) γ_1 is the straight line segment from 0 to $a + ib$ ($a, b \in \mathbb{R}$)
- (ii) γ_2 is the circle $|z| = R$
- (iii) γ_3 is the boundary of the square $[0, 1] \times [0, 1]$ with $\mathbb{C}$ considered as $\mathbb{R}^2$
- (iv) γ_4 is the ellipse $x^2/a^2 + y^2/b^2 = 1$
- (v) γ_5 is the line from 0 to $2i$ and then from $2i$ to $4 + 2i$
- (vi) γ_6 is the line segment from 0 to 1, 1 to $1 + i$ and then from $1 + i$ to 0
- (vii) γ_7 is given by $\gamma_7(t) = t + it^2$ on $[0, 1]$.





(i) γ_1 may be parameterized by $\gamma_1(t) = (a + ib)t$, $0 \leq t \leq 1$. Clearly,

$$I_1 = \int_0^1 [\operatorname{Re} \gamma_1(t)] \gamma_1'(t) dt = \int_0^1 ta(a + ib) dt = \frac{a(a + ib)}{2}.$$

(ii) Parameterizing γ_2 by $\gamma_2(t) = Re^{it}$, $0 \leq t \leq 2\pi$, we have

$$\begin{aligned} I_2 &= \int_0^{2\pi} (R \cos t)(iRe^{it}) dt \\ &= iR^2 \int_0^{2\pi} [\cos^2 t + i \sin t \cos t] dt \\ &= iR^2 \left[\int_0^{2\pi} \frac{1 + \cos 2t}{2} dt + \frac{i}{2} \int_0^{2\pi} \sin 2t dt \right] \\ &= iR^2 \left[\frac{1}{2} \left(t + \frac{\sin 2t}{2} \right) \Big|_0^{2\pi} + \frac{i}{2} \left(-\frac{\cos 2t}{2} \right) \Big|_0^{2\pi} \right] \\ &= iR^2 \pi. \end{aligned}$$





(iii) In this case, we have (see Figure 4.5)

$$I_3 = \left(\int_O^A + \int_A^B + \int_B^C + \int_C^O \right) x \, dz.$$

For the sake of convenience, we parameterize OA, AB, BC and OC , as follows:

$$C_1(t) = (1-t) \cdot 0 + t \cdot 1 = t$$

$$C_2(t) = (1-t) \cdot 1 + t \cdot (1+i) = 1 + ti$$

$$C_3(t) = (1-t) \cdot (1+i) + t \cdot i = 1 - t + i$$

$$C_4(t) = (1-t) \cdot i + t \cdot 0 = (1-t)i,$$





where $t \in [0, 1]$ and $\gamma_3 = C_1 + C_2 + C_3 + C_4$ (see Figure 4.5). Utilizing these, i.e. without reparameterizing, we have

$$I_3 = \sum_{k=1}^4 \int_0^1 (\operatorname{Re} C_k(t)) C'_k(t) dt$$

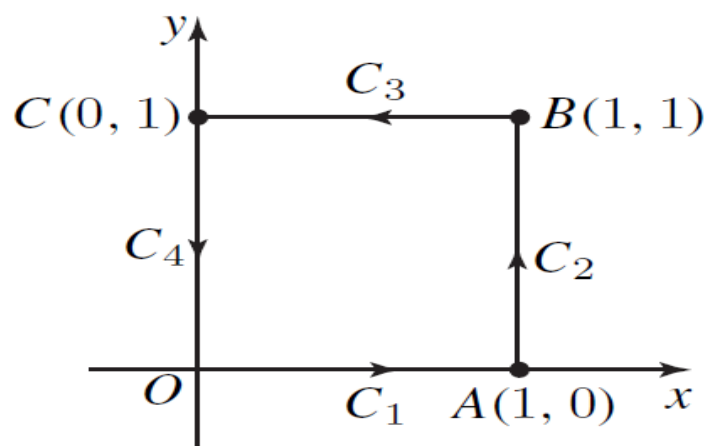


Figure 4.5: Curve $\gamma_3 = C_1 + C_2 + C_3 + C_4$.



$$\begin{aligned} &= \int_0^1 t \, dt + \int_0^1 1 \cdot i \, dt + \int_0^1 (1-t)(-1) \, dt + \int_0^1 0 \cdot (-i) \, dt \\ &= \frac{1}{2} + i - \frac{1}{2} + 0 = i. \end{aligned}$$

(iv) Write γ_4 as $\gamma_4(t) = a \cos t + ib \sin t$, $0 \leq t \leq 2\pi$. Then,

$$\begin{aligned} I_4 &= \int_0^{2\pi} (a \cos t)(-a \sin t + ib \cos t) \, dt \\ &= -\frac{a^2}{2} \int_0^{2\pi} \sin 2t \, dt + iab \int_0^{2\pi} \cos^2 t \, dt = iab\pi. \end{aligned}$$

The cases (v), (vi) and (vii) may be evaluated similarly and are left as a simple exercise. ●



Cauchy–Goursat Theorem



Statement

If $f(z)$ is analytic in a simply connected domain D , then for every closed contour C in D ,

$$\oint_C f(z) dz = 0.$$

Important Consequence

Contour integral depends only on endpoints in analytic region.

Example

Let $f(z) = e^z$. Then for any closed contour C ,

$$\oint_C e^z dz = 0.$$





Evaluate

$$\oint_C (z^3 + 2z + 1) dz$$

where C is the circle $|z| = 2$.

Solution

The function

$$f(z) = z^3 + 2z + 1$$

is a **polynomial**, and every polynomial is **analytic everywhere** in the complex plane (entire function).





Thus $f(z)$ is analytic inside and on the circle $|z| = 2$.

By Cauchy–Goursat theorem

$$\oint_C f(z)dz = 0$$

Final Answer

0



Cauchy Integral Formula



If f is analytic inside and on a simple closed contour C , and a lies inside C , then

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - a} dz.$$

Derivative Form

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - a)^{n+1}} dz.$$



Example



Evaluate

$$\oint_C \frac{e^z}{z-1} dz$$

where $C : |z| = 2$.

Since $z = 1$ lies inside:

$$= 2\pi i \cdot e^1 = 2\pi i e.$$





$$(a) \int_{|z|=1} \frac{\sin z}{z} dz = 2\pi i (\sin z) \Big|_{z=0} = 0$$

$$(b) \int_{|z|=1} \frac{\cos z}{z(z-4)} dz = 2\pi i \frac{\cos z}{z-4} \Big|_{z=0} = -\frac{\pi}{2}i$$

$$(c) \int_{|z-a|=1} \frac{e^{2\pi z}}{z-a} dz = 2\pi i e^{2\pi a}$$

$$(d) \int_{|z|=2} \frac{e^z + z^2}{z-1} dz = 2\pi i [e + 1]$$





(e) If $f(z) = \sin(\pi z^2) + \cos(\pi z^2)$, then by Cauchy's deformation of contour

$$\begin{aligned}\int_{|z|=4} \frac{f(z)}{(z-2)(z-3)} dz &= \int_{|z-2|=1/2} \frac{f(z)/(z-3)}{z-2} dz \\ &\quad + \int_{|z-3|=1/2} \frac{f(z)/(z-2)}{z-3} dz \\ &= 2\pi i(-f(2) + f(3)) = 0.\end{aligned}$$





(f) By Cauchy's deformation of contour

$$\begin{aligned}\int_{|z|=4} \frac{e^z}{z(z-1)} dz &= \int_{|z|=1/2} \frac{e^z/(z-1)}{z} dz + \int_{|z-1|=1/2} \frac{e^z/z}{z-1} dz \\ &= 2\pi i(-1 + e).\end{aligned}$$

$$(g) \int_{|z|=1} \frac{\cos(e^z)}{z} dz = 2\pi i \cos 1 = \pi i(e^i + e^{-i}).$$

•



Liouville's Theorem



Statement

Every bounded entire function is constant.

Application

Proves Fundamental Theorem of Algebra.

Example

If $|f(z)| \leq 5$ for all $z \in \mathbb{C}$, then
 $f(z) = \text{constant}$.





Show that e^z is not bounded

Consider

$$f(z) = e^z$$

Check boundedness

Let $z = x + iy$

$$\begin{aligned}|e^z| &= |e^{x+iy}| \\ &= |e^x(\cos y + i \sin y)| \\ &= e^x\end{aligned}$$

As $x \rightarrow \infty$,

$$|e^z| \rightarrow \infty$$





Entire function bounded by polynomial

Let $f(z)$ be entire and

$$|f(z)| \leq 3|z| + 2$$

for all z .

Idea

Consider

$$g(z) = \frac{f(z)}{z}$$





For large z ,

$$|g(z)| \leq 3 + \frac{2}{|z|}$$

Thus $g(z)$ is bounded.

By Liouville's theorem

$$g(z) = c$$

Hence

$$f(z) = cz$$

Thus $f(z)$ is linear.



Maximum Modulus Theorem



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If f is analytic and non-constant in a domain D , then
 $|f(z)|$ cannot attain its maximum inside D .

Maximum occurs only on boundary.



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Example 1: Maximum of $|z|$ on a closed disk

Find the maximum value of

$$|f(z)| = |z|$$

in the region $|z| \leq 2$.

Solution

The function

$$f(z) = z$$

is analytic everywhere.





The modulus is

$$|f(z)| = |z|$$

Inside the disk $|z| \leq 2$

$$|z| \leq 2$$

Thus

$$|f(z)| \leq 2$$

The maximum value 2 occurs on the boundary

$$|z| = 2$$





Therefore

$$\max |f(z)| = 2$$

Conclusion

The maximum occurs on the boundary, verifying the Maximum Modulus Theorem.

Example 2: Polynomial Function

Find the maximum of

$$|f(z)| = |z^2|$$

in the disk $|z| \leq 1$.





Solution

$$f(z) = z^2$$

Since

$$|z^2| = |z|^2$$

Inside the disk

$$|z| \leq 1$$

Thus

$$|z^2| \leq 1$$





Maximum occurs when

$$|z| = 1$$

Hence

$$\max |f(z)| = 1$$

This occurs on the boundary circle.



Taylor Series



If $f(z)$ analytic in disk $|z - a| < R$,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z - a)^n.$$

Example

Expand e^z about $z = 0$:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$



Example



Expand $\sin z$ about $z = 0$

Function

$$f(z) = \sin z$$

Derivatives:

$$f'(z) = \cos z$$

$$f''(z) = -\sin z$$

$$f'''(z) = -\cos z$$





At $z = 0$

$$\sin 0 = 0, \quad \cos 0 = 1$$

Thus

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots$$

Example. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \frac{1}{1+x^4}.$$



Zeros of Analytic Functions



If $f(a) = 0$ and

$$f(z) = (z - a)^m g(z),$$

where $g(a) \neq 0$,

then a is zero of order m .

Example

$$f(z) = z^3(z - 2)$$

Zero at 0 of order 3.





This function admits a real power series about any point $a \in \mathbb{R}$. Yet the power series about $a = 0$, given by

$$f(x) = \sum_{n=0}^{\infty} (-1)^n x^{4n}$$

has $(-1, 1)$ as its interval of convergence. On the other hand, its complex analog

$$f(z) = \frac{1}{1 + z^4}$$

has singularities at $z_k = e^{i\pi(1+2k)/4}$, $k = 0, 1, 2, 3$. Clearly, the distance from 0 to the nearest singularity is 1, which necessarily is the radius of convergence for the corresponding series about 0. ●



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Isolated and Non-isolated Singularities

A point $z = a$ is called a *regular point* for a complex-valued function f if f is analytic at a . Point a is called a *singular point* or a *singularity*, of f , if f is not analytic at a but every neighborhood of a contains at least one point at which f is analytic. A singular point a is said to be an *isolated singular point* or an *isolated singularity* of f if f is analytic in some deleted neighborhood of a . Otherwise, we say a is a *non-isolated singular*



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Types of Singularities



- (1) *Removable singularity*, which upon closer examination reveals that this is not actually considered to be a singular point at all. More precisely, an isolated singularity $z = a$ of f is said to be a removable singularity for f if $\lim_{z \rightarrow a} f(z)$ exists in $\mathbb{C}$.
- (2) *Pole* arises from the reciprocal of an analytic function with zero. More precisely, an isolated singularity $z = a$ of f is said to be a pole for f if $\lim_{z \rightarrow a} f(z) = \infty$. (Note that $f(z)$ is defined if z is near enough to a and $z \neq a$.)
- (3) *Essential singularity*, which is neither a removable singularity nor a pole. More precisely, an isolated singularity $z = a$ of f is said to be an essential singularity for f if $\lim_{z \rightarrow a} f(z)$ does not exist in $\mathbb{C}_\infty$. The behavior of a function in a neighborhood of an essential singularity is described by the *Casorati-Weierstrass theorem* ().

Removable Singularity



An isolated singularity z_0 of f is called *removable* or that f has a *removable singularity* at z_0 if f can be defined at z_0 so that it becomes analytic at z_0 .

Example. Consider the following functions:

$$\begin{aligned} f_1(z) &= \sin z \quad (z \neq 1), & f_4(z) &= \left(\frac{e^z - 1}{z} \right)^2 \quad (z \neq 0) \\ f_2(z) &= \frac{z}{e^z - 1} \quad (z \neq 0), & f_5(z) &= \frac{2(1 - \cos z)}{z^2} \quad (z \neq 0) \\ f_3(z) &= \frac{\sin z}{z} \quad (z \neq 0), & f_6(z) &= \frac{-\operatorname{Log}(1 - z)}{z} \quad (z \neq 0). \end{aligned}$$

All of these functions except f_1 have removable singularities at 0; and f_1 has a removable singularity at $z = 1$. These singularities can be removed by letting $f_1(1) = \sin 1$ and $f_j(0) = 1$ ($2 \leq j \leq 6$), respectively. ●



We have seen that if f is bounded in a deleted neighborhood of an isolated singularity at z_0 , then z_0 is a removable singularity of f . Thus if z_0 is not a removable singularity of f then, f is not bounded near z_0 . We might then ask whether $(z - z_0)^n f(z)$ is bounded near z_0 for some $n \in \mathbb{N}$. In this case, we say that the point z_0 is a pole of f and the smallest positive integer n such that $(z - z_0)^n f(z)$ is bounded near z_0 is called the *order of the pole* at z_0 . A pole of order one is called a *simple pole*. For example, the function $f(z) = z^8/(z - 1)^2$ has a double pole at $z = 1$.



Essential Singularity



Let $f(z)$ be analytic in a punctured neighborhood of $z = a$ (i.e., analytic for $0 < |z - a| < r$).

If the **Laurent series** of $f(z)$ about $z = a$

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - a)^n$$

contains **infinitely many negative power terms**, then $z = a$ is called an **essential singularity**.

Equivalently, the **principal part** of the Laurent series has infinitely many terms.



Example



$$f(z) = \sin\left(\frac{1}{z}\right)$$

Find the singularity at $z = 0$.

Solution

Using the expansion

$$\sin w = w - \frac{w^3}{3!} + \frac{w^5}{5!} - \dots$$

Substitute $w = \frac{1}{z}$

$$\sin\left(\frac{1}{z}\right) = \frac{1}{z} - \frac{1}{3!z^3} + \frac{1}{5!z^5} - \dots$$

This has infinitely many negative powers.

Hence,

$$z = 0$$

is an **essential singularity**.



Laurent Series



Definition

If a function $f(z)$ is analytic in an **annulus**

$$r < |z - a| < R$$

then it can be expanded as a **Laurent series**

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - a)^n$$

or

$$f(z) = \cdots + \frac{a_{-2}}{(z - a)^2} + \frac{a_{-1}}{z - a} + a_0 + a_1(z - a) + a_2(z - a)^2 + \cdots$$





The part

$$\frac{a_{-2}}{(z-a)^2} + \frac{a_{-1}}{z-a}$$

is called the **principal part**.

Example 1

Expand $\frac{1}{z(1-z)}$ about $z = 0$

Solution

First rewrite

$$\frac{1}{z(1-z)} = \frac{1}{z} \cdot \frac{1}{1-z}$$

Using the geometric series

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots$$

for $|z| < 1$





Multiply by $\frac{1}{z}$:

$$\frac{1}{z(1-z)} = \frac{1}{z} + 1 + z + z^2 + z^3 + \dots$$

Laurent Series

$$\frac{1}{z} + 1 + z + z^2 + z^3 + \dots$$

Example 2

Expand $\frac{1}{z(z-1)}$ about $z = 0$



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Example 3

Expand $\frac{1}{z^2(1-z)}$ about $z = 0$

Example 4

Expand $\frac{1}{z-2}$ about $z = 0$ for $|z| < 2$

Example 5

Expand $\frac{1}{z-1}$ about $z = 0$ for $|z| < 1$

Residues



The coefficient a_{-1} in Laurent series is residue.

$$\text{Res}_{z=a} f(z) = a_{-1}$$

Simple Pole Formula

$$\text{Res}_{z=a} f(z) = \lim_{z \rightarrow a} (z - a) f(z)$$



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Example 1

Find the residue of

$$f(z) = \frac{e^z}{z-1}$$

at $z = 1$.

Solution

Since $z = 1$ is a simple pole,

$$\begin{aligned}\operatorname{Res}_{z=1} f(z) &= \lim_{z \rightarrow 1} (z-1) \frac{e^z}{z-1} \\ &= e^1\end{aligned}$$





Example 2

Find the residue of

$$f(z) = \frac{1}{z(z-2)}$$

at $z = 0$.

Solution

$$\begin{aligned}\text{Res}_{z=0} f(z) &= \lim_{z \rightarrow 0} z \frac{1}{z(z-2)} \\ &= \lim_{z \rightarrow 0} \frac{1}{z-2}\end{aligned}$$





Example 3

Find the residues of

$$f(z) = \frac{1}{z^2 - 1}$$

Step 1: Factor

$$z^2 - 1 = (z - 1)(z + 1)$$

Poles at

$$z = 1, \quad z = -1$$

Residue at $z = 1$





$$\begin{aligned}\text{Res}_{z=1} &= \lim_{z \rightarrow 1} (z-1) \frac{1}{(z-1)(z+1)} \\ &= \frac{1}{2}\end{aligned}$$

Residue at $z = -1$

$$\begin{aligned}\text{Res}_{z=-1} &= \lim_{z \rightarrow -1} (z+1) \frac{1}{(z-1)(z+1)} \\ &= -\frac{1}{2}\end{aligned}$$

Example (Residue using Derivative Formula)

If $z = a$ is a pole of order 2

$$\text{Res}_{z=a} f(z) = \lim_{z \rightarrow a} \frac{d}{dz} [(z-a)^2 f(z)]$$



Find residue of

$$f(z) = \frac{1}{z^2(z-1)}$$

at $z = 0$.

Since pole of order 2:

$$\text{Res}_{z=0} = \lim_{z \rightarrow 0} \frac{d}{dz} \left[z^2 \frac{1}{z^2(z-1)} \right]$$

$$= \lim_{z \rightarrow 0} \frac{d}{dz} \left(\frac{1}{z-1} \right)$$

$$= -\frac{1}{(z-1)^2}$$

At $z = 0$

$$= -1$$



Cauchy Residue Theorem



If f analytic except isolated singularities inside C ,

$$\oint_C f(z)dz = 2\pi i \sum \text{Residues inside.}$$

Example 1

Evaluate

$$\oint_{|z|=2} \frac{1}{z-1} dz$$

Solution

Singularity:

$$z = 1$$

Since





$$|1| < 2$$

the pole lies inside the contour.

Residue at $z = 1$

$$\text{Res}_{z=1} \frac{1}{z-1} = 1$$

Thus

$$\begin{aligned} \oint_{|z|=2} \frac{1}{z-1} dz &= 2\pi i(1) \\ &= 2\pi i \end{aligned}$$





Example 2

Evaluate

$$\oint_{|z|=3} \frac{z}{z^2 + 1} dz$$

Solution

Factor denominator

$$z^2 + 1 = (z - i)(z + i)$$

Singularities

$$z = i, \quad z = -i$$





Both lie inside $|z| = 3$.

Residue at $z = i$

$$\begin{aligned}\text{Res}_{z=i} &= \lim_{z \rightarrow i} (z - i) \frac{z}{(z - i)(z + i)} \\ &= \frac{i}{2i} = \frac{1}{2}\end{aligned}$$

Residue at $z = -i$

$$\text{Res}_{z=-i} = \lim_{z \rightarrow -i} (z + i) \frac{z}{(z - i)(z + i)}$$





$$= \frac{-i}{-2i} = \frac{1}{2}$$

Sum of residues

$$\frac{1}{2} + \frac{1}{2} = 1$$

Thus

$$\begin{aligned} \oint_{|z|=3} \frac{z}{z^2+1} dz &= 2\pi i(1) \\ &= 2\pi i \end{aligned}$$





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